

Trees and Binary Tree

11 May 2026 19:58

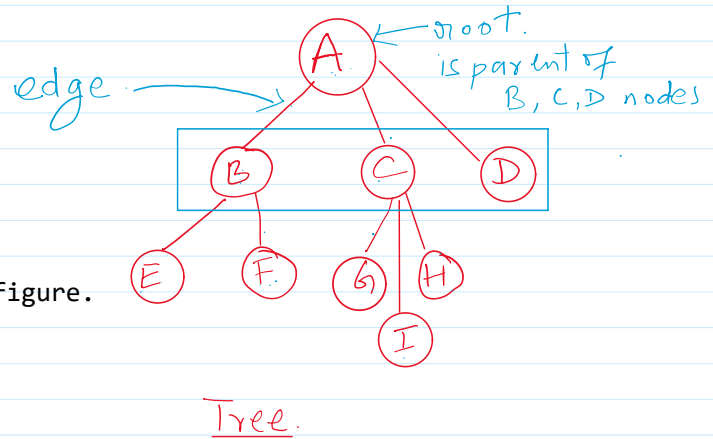
What is Tree in Computer Science?

A **Tree** is a non-linear data structure used to store hierarchical data.

Unlike arrays or linked list, tree represent data in a parent-child relationship.

Example:

- File systems
- Organizational hierarchy
- XML/HTML structure
- Database indexing



Root Node: The topmost node of a tree.

Example: A is the root node in the given figure.

Parent Node: A node having child nodes

Example: A is a parent of B, C, D.

Child Node: Nodes connected below a parent node.

Example: B, C, D are child nodes of A.

Leaf node: Nodes having no children

Example: E, F, G, I, H, D in our diagram.

Siblings: Node having same parent.

Example: B, C, D are siblings having A as parent.

Edge: Connection between two nodes

Example: A-----B

Height of Tree: Maximum number of edges from root to deepest node.

Example: Height = 2 meaning Maximum number of edges from root to deepest node in our diagram.

Types of Trees

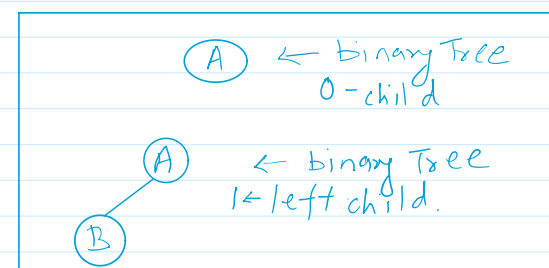
- General Tree
- Binary Tree
- Binary Search Tree
- AVL tree
- Heap
- B-Tree
- Expression Tree
- Recursion Tree
- Red-Black Tree
- And many more.

Binary Tree

A **Binary Tree** is a tree in which each node can have at most:

- One left child
- One Right child
- Or No-child

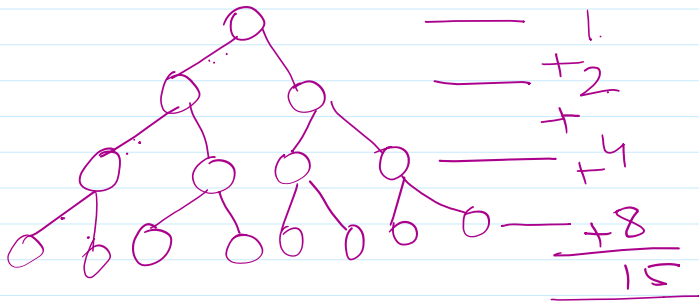
Properties of Binary Tree.



Properties of Binary Tree.

⊗ Maximum nodes at level $L = 2^L$

⊗ Maximum ^{number of} nodes in tree of height h .



$$= 2^{h+1} - 1 \quad [\because h = \text{height}]$$

$$= 2^{3+1} - 1 = 16 - 1 = \underline{15}$$

⊗ Minimum height of binary tree with $n = 15$ node.

$$2^{h+1} - 1 = 15 \quad \text{find } h.$$

Handwritten calculation for finding the minimum height of a binary tree with 15 nodes.

$$2^{h+1} - 1 = 15$$

$$2^{h+1} = 16$$

$$2^{h+1} = 2^4$$

$$h+1 = 4$$

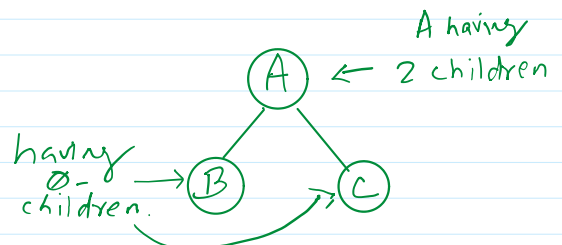
$$h = 4 - 1 = 3$$

$$h_{\min} = \log_2(n+1) - 1$$

Types of Binary Trees

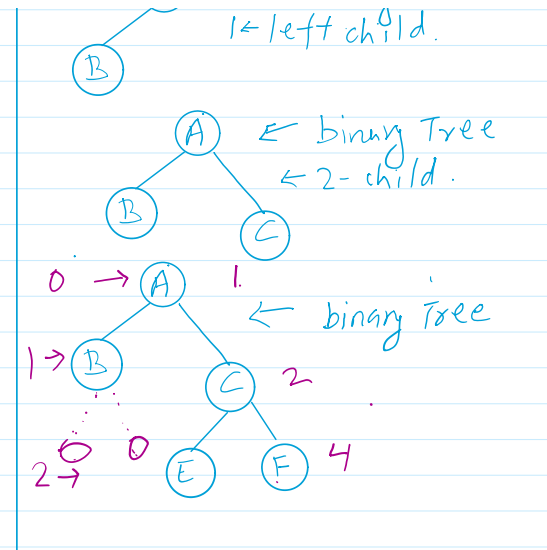
1. Full Binary Tree: Every node has either:

- 0-children
- Or 2-children

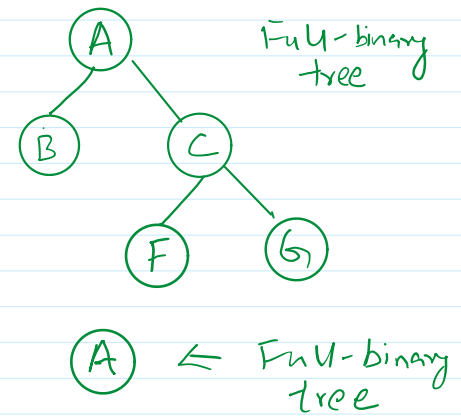
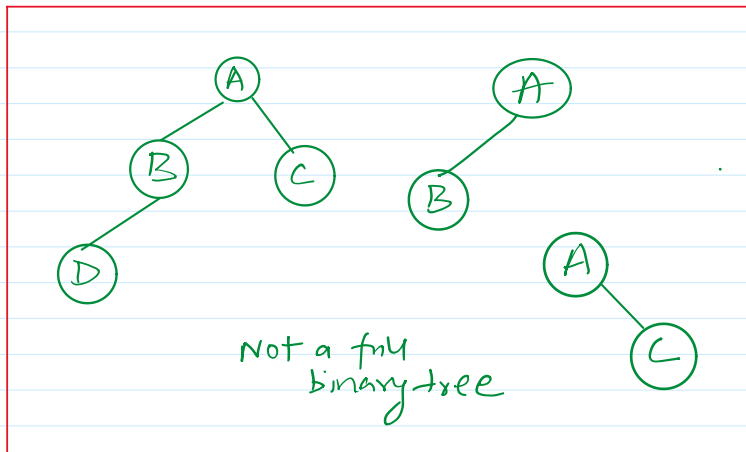


(A)

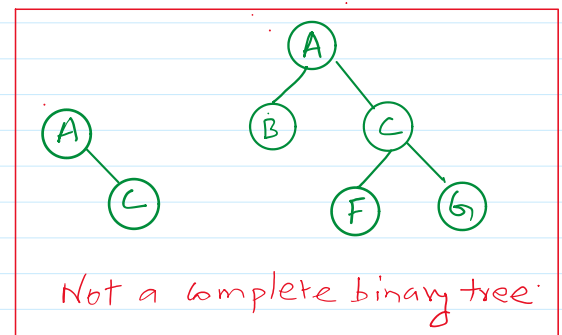
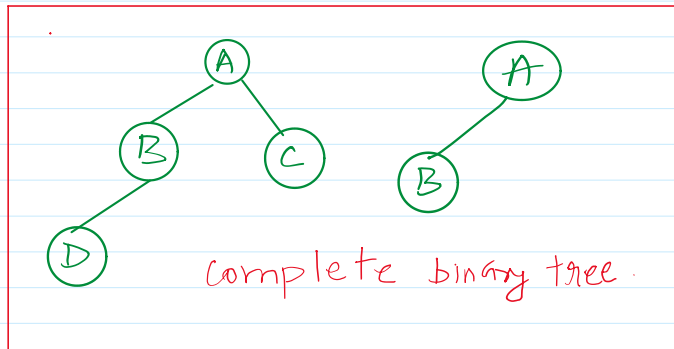
Full binary



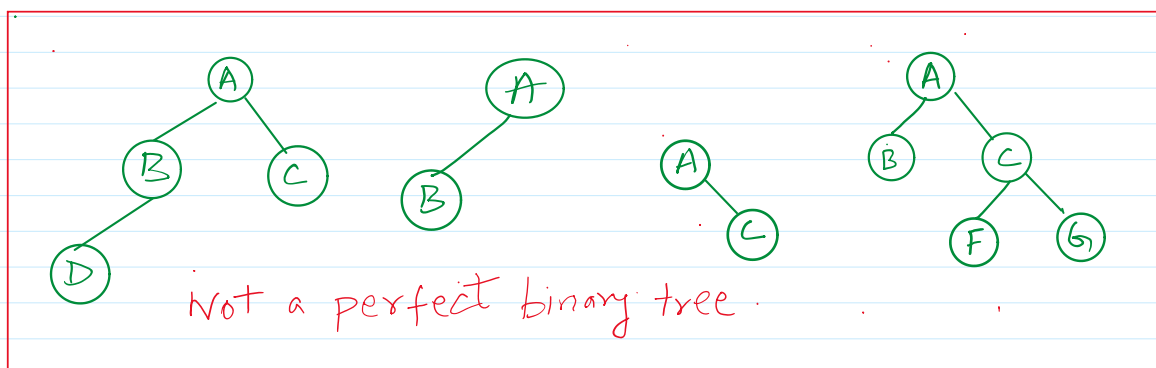
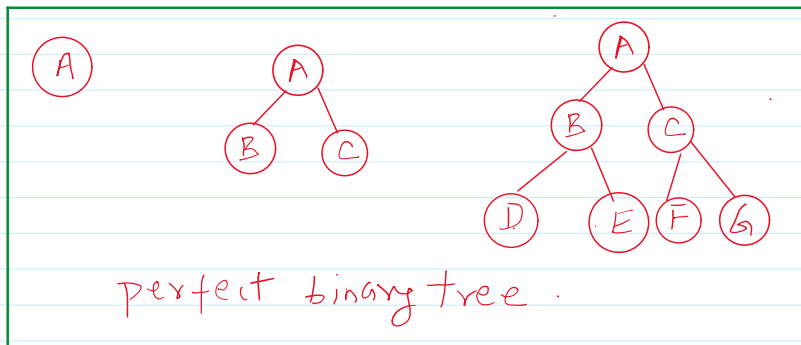
Structures of Binary Tree.



2. **Complete Binary Tree:** All levels completely filled except possibly last level. Last level filled from left to right.



3. **Perfect Binary Tree:** All internal nodes have 2 children and all leaf nodes are at same level.



4. Degenerated Binary Tree: Each parent has only one child, it looks like linked list.

